ADSDB P1 · UPC

EcoSentinel

Wildlife data connected to its sources

Mario Rodrigo
Sambhav · Wassim



Photo: lynxexsitu.es · CC BY 3.0 es

One wildlife question, several sources

“I have a photograph. What can help me understand it?”

Find related images
Listen to recordings
Explore descriptions

Proposed use: education and biodiversity exploration

The reference corpus

198

accepted assets
10 Iberian wildlife species

Modality	Assets	Source
Images	100	Wikimedia
Audio	34	Xeno-canto
Text	64	Wikipedia / GBIF

Uneven availability: no audio for the ocellated lizard

Four zones, four contracts

Zone	Responsibility	Storage
Landing	Preserve originals	S3 · Docker
Formatted	Standardise representation	S3 · Docker
Trusted	Assess quality and provenance	S3 · Docker
Exploitation	Prepare vector retrieval	Persistent Chroma

Queries use an independent snapshot of the index

Preserving originals, standardising formats

Preserved original

Temporal and Persistent Landing
Copy verified with SHA-256
Content identity

Common representation

Image: RGB JPEG
Audio: FLAC PCM16
Text: UTF-8, NFC and LF

Model-specific segmentation and resampling happen later

Trusted retains potentially useful signals

198 / 198

accepted under the corpus technical policies

32

possible watermark flags

9

low-energy audio warnings

Weak signals remain available for review

198 assets, 6,021 embeddings

6,021

persisted vectors
Segments with identity, hash and contract

Collection	Vectors
Image · CLIP	100
Audio · CLAP	357
Text · E5	547
Image + text · CLIP	1,763
Audio + text · CLAP	3,254

The encoder follows the relationship

Encoder	Represented relationship	Practical choice
CLIP	Aligned image and text	Visual concepts
CLAP	Aligned audio and text	Sound descriptions
E5 small	Multilingual text	Compact representation

CPU · pretrained models · separate contracts

No comparative model benchmark for this domain

Task 1: similarity within a modality

Query	Results
Image	Other images
Audio	Other recordings
Text	Other texts

One result per asset

Entire reference asset excluded

External files serve as queries and are not indexed

Task 2: retrieval across modalities

“Iberian lynx in a forest”

CLIP: image and text
CLAP: audio and text

All modalities:
rank fusion with RRF



Actual combined search interface (Spanish)

Task 3: evidence before generation

Question + optional image

01 Retrieval with E5 / CLIP

02 Text context and BLIP observation

03 Qwen 2.5 · 3B · GGUF · CPU

04 Checked quotations and explicit status

Qwen receives text. BLIP interprets the image.

An answer with sources you can inspect

Corpus candidate: Pelophylax perezi

Habitat [S1] “areas with little current”

Diet [S2] “They feed on terrestrial and aquatic prey”

Translated extracts from a real run · the candidate does not confirm the photo’s species

The second run matters too

16 steps

Execution lock
Per-step report and log
Explicit failure and retry

7

vectors in the first run

0

new in the second run

Isolated run with three synthetic assets and CPU models

What we validated

344

offline tests

+ 1 real integration test

71.25%

combined coverage

6,021 original vectors
with identical files and hashes

Windows / Linux CI configured · verify the published commit

A query with a traceable path

Preserved originals
Retrieval across three modalities
Answers linked to sources

Next evaluation: relevance and semantic quality with labelled examples and specialist review